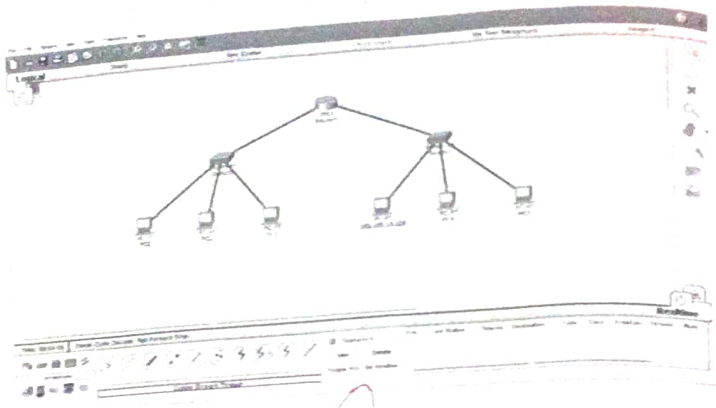


EX.NO: 09

DATE: 16/09/2021

IMPLEMENTATION OF SUBNETTING IN CISCO PACKET TRACER SIMULATOR.

AIM :-

To implement classless IP subnetting and configure the subnetted network using cisco packet Tracer.

REQUIREMENTS :-

- → Cisco Packet Tracer
- ⇒ Routers
- ⇒ Switches
- ⇒ PCs
- ⇒ Ethernet cables

PROCEDURE :-

1) open cisco packet tracer and create a new network topology.

2) Add routers, switches & PCs to the workspace.

3) Connect the devices using appropriate cables

4) Use the network 192.168.1.0/24 & subnet it using a /27 subnet mask.

5) The /27 mask provides 8 subnets with 30 usable host addresses per subnet.

6) Configure the router interfaces with suitable IP addresses & subnets masks.

7) Configure the switch ports in access mode.

8) Configure each PC with its IP address, subnet mask, default gateway & DNS information.

Input :-

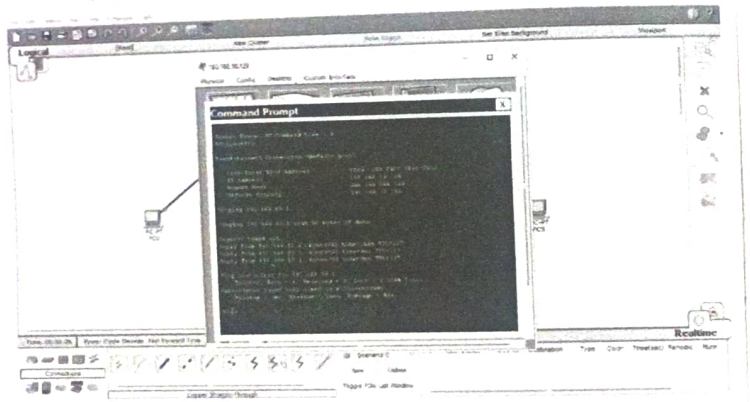
Network address :- 192.168.1.0/24

Subnet mask :- /27

IP addresses are assigned to routers & PCs are given in manual.

output :-

The subnetted network topology is successfully created & configured in cisco packet tracer. The PCs & router interfaces are assigned appropriate IP addresses.



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC4(2)	Router2	ICMP		0.000	N	12
	Successful	PC4(2)	PC2(1)(1)	ICMP		0.000	N	13
	Successful	PC0	Router0	ICMP		0.000	N	14

9) Enable router interfaces using 'no shutdown' command.

10) Test connectivity between PCs & b/w the router & PCs using 'ping' command.

PROGRAM :-

Router configuration :-

```
enable
configure terminal
interface Fast Ethernet 0/0
ip address 192.168.1.1
255.255.255.224
no shutdown
exit
interface FastEthernet 0/1
ip address 192.168.2.1
255.255.255.224
no shutdown
exit
```

Switch configuration :-

```
enable
configure terminal
interface Fast Ethernet 0/1
switch port mode access
exit
interface Fast Ethernet 0/2
switchport mode access
exit
```

RESULT :-

Thus, the network was successfully subnetted & configured in Cisco packet tracer & the connectivity b/w the network devices are verified.